Multivariate Analysis of the Physicochemical Characteristics of Red and White Wines

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Introduction

Wine quality is influenced by various physicochemical characteristics, including acidity, residual sugar, chlorides, sulfur dioxide, density, and alcohol content. These properties can be measured through laboratory tests and may help describe differences in the chemical composition of wine. Understanding how these characteristics vary and relate to one another is important when studying wine quality and comparing different types of wine.

The UCI Wine Quality Dataset contains physicochemical measurements and quality ratings for red and white wines. Since these measurements may be correlated, analyzing each variable separately may not fully explain the overall patterns in the data. Multivariate statistical methods provide a way to examine several variables simultaneously, identify major patterns of variation, and explore relationships among chemical characteristics.

This study applies exploratory data analysis (EDA), multivariate analysis of variance (MANOVA), principal component analysis (PCA), and exploratory factor analysis (EFA) to the Wine Quality Dataset. EDA is used to summarize the data and examine distributions, correlations, and potential outliers. MANOVA investigates whether red and white wines differ in their combined mean values for selected physicochemical characteristics. PCA is used to reduce the dimensionality of the eleven physicochemical variables while retaining as much information as possible, while EFA explores whether underlying factors can explain patterns of correlation among these variables.

Overall, the study aims to compare the physicochemical profiles of red and white wines and determine whether the chemical measurements can be summarized into a smaller number of meaningful components or factors. The findings are interpreted alongside the assumptions and limitations of each statistical method.

Dataset

This study uses the Wine Quality Dataset from the UCI Machine Learning Repository, originally presented by Cortez et al. (2009). The dataset contains physicochemical measurements and quality ratings for red and white vinho verde wines from northern Portugal. It is designed to support the analysis of relationships between wine properties and quality.

The combined dataset contains 6,497 wine samples, consisting of 1,599 red wine samples and 4,898 white wine samples. Each observation represents a wine sample. The dataset includes eleven physicochemical variables: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. It also contains a quality score, which represents the wine's quality rating, and a wine color label identifying each sample as red or white.

The physicochemical variables describe different aspects of wine composition. For example, acidity-related variables measure aspects of the wine's acid content, residual sugar measures the amount of sugar remaining after fermentation, sulfur dioxide variables describe sulfur dioxide levels, and alcohol measures alcohol content. Density and the other chemical measurements provide additional information about the overall composition of the samples.

For this analysis, the red and white wine datasets were combined into a single dataset, and a wine color variable was used to distinguish the two groups. Exploratory data analysis was conducted to examine the structure of the data, descriptive statistics, variable distributions, correlations, and potential outliers. The eleven physicochemical variables were used in PCA and EFA. MANOVA was conducted using five selected variables: volatile acidity, residual sugar, chlorides, density, and alcohol. The quality score and wine color label were not included among the eleven physicochemical variables used for PCA and EFA.

The dataset was obtained from the UCI Machine Learning Repository. Its use allows the study to compare the chemical profiles of red and white wines and investigate patterns of variation and correlation among their physicochemical characteristics.

Research Questions and Hypotheses

This study addresses three research questions using MANOVA, principal component analysis (PCA), and exploratory factor analysis (EFA). These questions focus on differences between red and white wines and the relationships among their physicochemical characteristics.

Research Question 1: Differences Between Red and White Wines

Research question: Do red and white wines differ significantly in their combined physicochemical characteristics?

MANOVA is used to examine differences between red and white wines based on five selected variables: volatile acidity, residual sugar, chlorides, density, and alcohol.

Null hypothesis (H0): Red and white wines have equal population mean vectors for the five selected physicochemical variables.

Alternative hypothesis (H1): Red and white wines have different population mean vectors for the five selected physicochemical variables.

Research Question 2: Dimensionality Reduction Using PCA

Research question: Can the eleven physicochemical variables be summarized into a smaller number of principal components while retaining a substantial proportion of the total variance?

PCA is used to identify components that summarize the variation across the eleven physicochemical measurements. The number of components is evaluated using eigenvalues, a scree plot, explained variance, cumulative explained variance, and parallel analysis.

Since PCA is primarily a dimensionality reduction technique, this question is evaluated by examining how much variance the retained components explain and whether their loadings reveal interpretable patterns among the variables.

Research Question 3: Underlying Factors Among Physicochemical Variables

Research question: Can a smaller number of underlying factors explain the correlation patterns among the eleven physicochemical variables?

EFA is used to explore whether the observed correlations can be represented by a smaller set of underlying factors. The suitability of factor analysis is assessed using the Kaiser-Meyer-Olkin (KMO) measure, Bartlett's test of sphericity, parallel analysis, factor loadings, communalities, uniqueness values, and model-fit statistics.

Because EFA is exploratory, the analysis does not assume in advance that a reliable factor structure will be identified. The results are interpreted according to the adequacy of the diagnostics and the resulting factor solution.

Methodology

Data Preparation

The red and white wine datasets were combined into a single dataset, with a categorical variable named `wine_color` used to distinguish the two wine types. The combined dataset contained 6,497 observations, consisting of 1,599 red wines and 4,898 white wines. The data were examined to understand their structure, variable types, missing values, duplicate observations, and potential outliers before conducting the statistical analyses.

The eleven physicochemical variables used in the dimensionality reduction analyses were fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. The quality score and wine color label were excluded from PCA and EFA because these analyses focused on the relationships among the chemical measurements. For MANOVA, five variables were selected: volatile acidity, residual sugar, chlorides, density, and alcohol. These variables represent different aspects of wine composition and were selected to investigate differences between red and white wines.

For PCA, the physicochemical variables were centered and standardized to have comparable scales because they were measured in different units and had different ranges. For EFA, the relationships among the eleven variables were examined using their correlation matrix. Missing observations were handled through complete-case analysis where applicable. The number of observations retained for each analysis was checked to account for any observations excluded during preprocessing.

Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to summarize the dataset and identify important patterns before performing the multivariate analyses. Descriptive statistics, including means, medians, standard deviations, variances, minimums, maximums, and distribution measures, were examined for the physicochemical variables and quality scores.

Histograms and boxplots were used to inspect variable distributions, skewness, and potential outliers. Comparisons between red and white wines were made using grouped descriptive statistics and boxplots. A correlation matrix was also examined to identify relationships among the numeric variables and potential redundancy between strongly correlated measurements.

The findings from EDA were used to understand the characteristics of the dataset, guide the selection of variables for MANOVA, and inform the interpretation of the PCA and EFA results. Potential outliers and non-normal distributions were considered when interpreting the subsequent analyses.

Multivariate Analysis of Variance

Multivariate analysis of variance (MANOVA) was used to determine whether red and white wines differed in their combined mean values for five physicochemical variables: volatile acidity, residual sugar, chlorides, density, and alcohol. Wine color was treated as the categorical independent variable, while the five chemical measurements served as the dependent variables.

The null hypothesis stated that red and white wines had equal population mean vectors for the selected variables. The alternative hypothesis stated that their population mean vectors differed. Pillai's trace was used as the primary multivariate test statistic because it is generally considered relatively robust to certain violations of MANOVA assumptions.

The analysis considered the assumptions of independent observations, multivariate normality, absence of influential multivariate outliers, and homogeneity of covariance matrices. These assumptions were assessed using the available data diagnostics, with the limitations of the diagnostic procedures taken into account. Following the multivariate test, separate univariate ANOVAs were conducted for the five dependent variables to identify which variables contributed to the observed group differences. Partial eta-squared values were calculated to describe the effect sizes, and Holm adjustment was applied to the follow-up p-values to account for multiple comparisons.

Principal Component Analysis

Principal component analysis (PCA) was conducted to reduce the eleven physicochemical variables to a smaller number of components while retaining as much of the total variability as possible. The analysis excluded the quality score and wine color label so that the components represented patterns in the chemical measurements themselves.

Before PCA, the eleven variables were centered and standardized. PCA was then performed using the `prcomp()` function in R with centering and scaling enabled. The number of components to retain was evaluated using eigenvalues, a scree plot, the proportion and cumulative proportion of explained variance, and parallel analysis.

The component loadings were examined to identify the variables contributing most strongly to each retained component. Variables with larger absolute loadings were used to interpret the underlying patterns represented by the components. The retained components were then evaluated in terms of their interpretability and the amount of total variance they explained.

Exploratory Factor Analysis

Exploratory factor analysis (EFA) was performed to investigate whether a smaller number of underlying factors could explain the correlation patterns among the eleven physicochemical variables. The same eleven chemical measurements used in PCA were included, while quality and wine color were excluded.

The correlation matrix was examined before factor extraction. Sampling adequacy was evaluated using the Kaiser-Meyer-Olkin (KMO) measure, and Bartlett's test of sphericity was used to test whether the correlation matrix differed significantly from an identity matrix. Parallel analysis was used to help determine the number of factors to retain.

Factors were extracted using minimum residual estimation and an oblimin rotation was applied, allowing the factors to be correlated. Factor loadings, communalities, uniqueness values, factor correlations, and model-fit statistics were examined to assess the resulting solutions. Solutions showing inadmissible estimates, such as communalities greater than one or negative uniqueness values, were treated as problematic. The factor structure was therefore interpreted cautiously, with particular attention to the adequacy of the KMO measure and the overall model fit.

Statistical Software

All data preparation, exploratory analysis, and multivariate statistical analyses were conducted using R in the RStudio environment. R was used to organize and summarize the data, generate plots, perform MANOVA and follow-up ANOVAs, calculate PCA components, and conduct EFA and its diagnostic tests.

Functions from packages used in the analyses included `dplyr` and `tidyr` for data preparation, `psych` for factor-analysis diagnostics and parallel analysis, and other relevant functions for multivariate testing, visualization, and descriptive statistics. The specific functions and packages used are documented in the accompanying R code to support reproducibility.

Results

Exploratory Data Analysis

The combined Wine Quality Dataset contained 6,497 observations, consisting of 1,599 red wine samples and 4,898 white wine samples. Exploratory data analysis was conducted to examine the distributions, variability, potential outliers, and relationships among the variables.

The descriptive statistics and boxplots showed differences in the distributions of several physicochemical characteristics. Some chemical variables, particularly residual sugar, chlorides, sulfur dioxide measurements, and sulphates, exhibited right-skewed distributions. This suggests that many observations had relatively low values, while a smaller number of samples had substantially higher values. The boxplots also indicated potential outliers in several variables. These observations were retained for interpretation unless specifically excluded during preprocessing.

The correlation matrix revealed several notable relationships among the physicochemical variables. Free sulfur dioxide and total sulfur dioxide had a strong positive correlation of approximately 0.720. This indicates that wines with higher free sulfur dioxide levels tended to have higher total sulfur dioxide levels. Density and alcohol had a strong negative correlation of approximately -0.668, suggesting that samples with higher alcohol content tended to have lower density. Residual sugar and density had a moderate positive correlation of approximately 0.521, while fixed acidity and density had a moderate positive correlation of approximately 0.478. Volatile acidity and citric acid showed a moderate negative correlation of approximately -0.384.

Comparisons between red and white wines indicated differences in several physicochemical characteristics, although the distributions of the two groups also showed some overlap. The quality score distributions had similar median values, generally around 6, with potential outliers at both the lower and higher ends. These findings suggest that wine color is associated with differences in chemical composition, while quality scores alone do not fully distinguish the two groups.

Overall, EDA identified correlations that were relevant to the subsequent multivariate analyses. In particular, the strong relationship between the two sulfur dioxide measurements and the negative relationship between density and alcohol indicated that several chemical characteristics shared substantial variation. The skewed distributions and potential outliers were also considered when interpreting the MANOVA and factor analysis results.

Multivariate Analysis of Variance (MANOVA)

MANOVA was conducted to determine whether red and white wines differed in their combined mean values for five physicochemical variables: volatile acidity, residual sugar, chlorides, density, and alcohol.

The null hypothesis stated that the population mean vectors of red and white wines were equal for the five selected variables. The alternative hypothesis stated that the population mean vectors differed.

Using Pillai's trace, the MANOVA result was statistically significant, Pillai's trace = 0.78529, approximate $F(5, 5314) = 3887.2$, $p < .001$. Therefore, the null hypothesis was rejected. The results provide strong evidence that red and white wines differ in their combined mean values for the five selected physicochemical characteristics.

Follow-up univariate ANOVAs were conducted to examine the individual variables contributing to the multivariate difference. All five variables showed statistically significant differences between red and white wines after Holm adjustment, with adjusted p-values below .001.

The partial eta-squared effect sizes differed considerably across the variables. Volatile acidity had a partial eta-squared value of approximately 0.416, indicating a large group effect. Chlorides also showed a substantial effect, with partial eta-squared of approximately 0.250, followed by density at 0.184 and residual sugar at 0.108. Alcohol had a much smaller effect, with partial eta-squared of approximately 0.003.

These results indicate that the difference between red and white wines was not equally strong across all five characteristics. Volatile acidity, chlorides, density, and residual sugar showed comparatively larger group effects, while the effect for alcohol was much smaller despite being statistically significant. Because the sample size was large, statistical significance should be interpreted alongside effect sizes.

Principal Component Analysis (PCA)

PCA was conducted using the eleven physicochemical variables after centering and standardizing them. Standardization was necessary because the variables were measured on different scales. The quality score and wine color label were excluded from this analysis.

The eigenvalue criterion and parallel analysis both suggested retaining three principal components. The first component explained approximately 27.17% of the total variance, the second explained 22.50%, and the third explained 14.43%. Together, the three components explained approximately 64.10% of the total variance. The remaining 35.90% was distributed across the other components.

The first principal component showed relatively large loadings for total sulfur dioxide (0.823), free sulfur dioxide (0.736), volatile acidity (-0.674), chlorides (-0.552), sulphates (-0.549), residual sugar (0.514), and fixed acidity (-0.475). This component represents a contrast between sulfur dioxide and residual sugar measurements and several other chemical characteristics. It suggests that a substantial portion of the variability in the dataset is associated with differences in sulfur dioxide levels and broader chemical composition.

The second principal component was strongly associated with density (-0.889) and alcohol (0.732), with additional contributions from residual sugar (-0.579), fixed acidity (-0.482), and chlorides (-0.432). This component primarily represents a density-alcohol contrast, reflecting the strong negative relationship between these two variables observed during EDA.

The third principal component had large loadings for citric acid (-0.727), pH (0.585), and fixed acidity (-0.534). This pattern suggests an acidity-related dimension in the data, reflecting relationships among different measures of acidity.

The signs of principal component loadings indicate the direction of each variable's relationship with a component, but the overall signs of a component can be reversed without changing the PCA solution. Therefore, the interpretation focuses on the relative loading patterns rather than treating positive or negative signs as inherently meaningful.

Overall, PCA showed that the eleven physicochemical measurements could be summarized by three components explaining 64.10% of the total variance. This provides a useful reduction in dimensionality, although some variation remains outside the retained components.

Exploratory Factor Analysis (EFA)

EFA was conducted to investigate whether a smaller number of underlying factors could explain the correlation patterns among the eleven physicochemical variables. Before factor extraction, the suitability of the correlation matrix for factor analysis was examined using the Kaiser-Meyer-Olkin measure and Bartlett's test of sphericity.

The overall KMO measure was 0.40, indicating poor sampling adequacy for factor analysis. Bartlett's test of sphericity was statistically significant, chi-squared(55) = 29789.89, $p < .001$. This indicates that the correlation matrix differed significantly from an identity matrix and that relationships existed among the variables. However, a significant Bartlett's test alone does not establish that the data are suitable for a reliable factor solution.

Parallel analysis suggested retaining four factors. A four-factor model was therefore examined using minimum residual extraction and oblimin rotation, which allows factors to correlate. The solution showed some interpretable loading patterns. Free sulfur dioxide and total sulfur dioxide loaded strongly on the same factor, with loadings of approximately 0.773 and 0.849, respectively. Another factor was strongly associated with density, while additional patterns involved acidity-related variables, chlorides, and sulphates.

However, the four-factor solution produced an inadmissible estimate for density. Its communality was approximately 1.048, resulting in a negative uniqueness estimate of approximately -0.048. This is known as a Heywood case and indicates that the factor solution cannot be accepted as a fully valid representation of the data.

A three-factor solution was also examined for comparison. It showed broadly similar patterns, including a factor associated with sulfur dioxide measurements, a density-alcohol contrast, and an acidity-related pattern. However, the three-factor solution also produced a Heywood case for density, with a communality of approximately 1.042 and a negative uniqueness estimate of approximately -0.042.

The three-factor model also showed poor overall fit, with a Tucker-Lewis index (TLI) of approximately 0.382 and a root mean square error of approximation (RMSEA) of approximately 0.251. The root mean square residual (RMSR) was approximately 0.06. Taken together with the low KMO value, these diagnostics indicate that the attempted factor solutions did not provide an adequate and stable representation of the underlying correlation structure.

Consequently, the EFA results should be interpreted as exploratory rather than conclusive. Although some loading patterns were consistent with the correlations observed during EDA and PCA, the inadmissible density estimates and poor model fit prevent the factor structure from being considered reliable. Further investigation would be necessary before drawing firm conclusions about latent factors underlying the physicochemical measurements.

Integrated Discussion

The three multivariate analyses provide complementary insights into the physicochemical characteristics of red and white wines. EDA identified differences in distributions, potential outliers, and several relationships among the chemical measurements. MANOVA formally tested whether the two wine color groups differed in their combined mean values for five selected characteristics. PCA summarized the overall variation among the eleven chemical measurements, while EFA explored whether their correlations could be explained by a smaller number of underlying factors.

The MANOVA results demonstrated a statistically significant multivariate difference between red and white wines. The follow-up ANOVAs showed that all five selected variables differed significantly between the groups, although the effect sizes varied. Volatile acidity had the largest partial eta-squared value, followed by chlorides, density, and residual sugar. Alcohol had a much smaller effect size. These results suggest that the chemical profiles of red and white wines differ across multiple characteristics, with some variables showing more pronounced group differences than others. However, the direction of each difference should be established using the group means rather than inferred from statistical significance alone.

PCA provided a more compact representation of the eleven physicochemical variables. The first three principal components explained approximately 64.10% of the total variance. The first component was associated with sulfur dioxide measurements and a contrast involving other chemical characteristics. The second component primarily reflected the relationship between density and alcohol, while the third represented an acidity-related pattern. These findings were consistent with several relationships identified during EDA and demonstrated that a substantial portion of the data's variability could be summarized using fewer dimensions.

EFA had a more limited outcome. Although parallel analysis suggested four factors and some loading patterns were interpretable, the KMO measure was low and both the four-factor and three-factor solutions produced inadmissible estimates for density. The three-factor model also showed poor fit. Therefore, the EFA did not establish a reliable latent factor structure for the eleven physicochemical variables. This illustrates the

distinction between PCA and EFA: PCA summarizes total variance in the observed measurements, whereas EFA attempts to explain their shared correlations through underlying factors. A component structure that is useful for dimension reduction does not necessarily imply that a valid latent factor model exists.

Several limitations should be considered when interpreting the findings. First, the dataset is observational, so the identified relationships and group differences do not establish causal relationships. Second, the numbers of red and white wine samples are unequal, which should be considered when interpreting group comparisons. Third, some variables showed skewness and potential outliers, which may affect the assumptions of multivariate statistical methods. Finally, the low KMO measure and inadmissible EFA solutions limited the conclusions that could be drawn about underlying factors. These limitations suggest that the results should be interpreted in terms of the dataset and methods used, rather than generalized without qualification to all wines.

Overall, the analyses indicate that red and white wines have significantly different multivariate physicochemical profiles and that PCA can summarize important patterns of variation among the chemical measurements. EFA, however, did not produce an adequate factor solution in this analysis. Together, the findings demonstrate the value of using complementary multivariate techniques while carefully evaluating their assumptions and limitations.

8. Conclusion

This study used exploratory data analysis, MANOVA, PCA, and EFA to examine the physicochemical characteristics of red and white wines. MANOVA identified a statistically significant difference between the groups across five selected chemical variables, with the largest effect observed for volatile acidity and the smallest for alcohol. PCA reduced the eleven physicochemical variables to three principal components that explained approximately 64.10% of the total variance. These components reflected patterns associated with sulfur dioxide and chemical composition, density and alcohol, and acidity-related measurements. In contrast, EFA did not produce a reliable factor solution because of poor sampling adequacy, inadmissible estimates for density, and poor model fit. The findings show that multivariate methods can reveal meaningful differences and patterns in wine chemistry, but their conclusions must be evaluated in light of diagnostic results and methodological limitations.

References

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